

Study of Industrially Important Microbes

Aim

To study the microbes which are used in industrial processes.

Requirements

A compound microscope, permanent slides of the following bacteria:

1. (1) *Streptococcus thermophilus*
2. (2) *Lactobacillus acidophilus*
3. (3) *Saccharomyces cerevisiae*
4. (4) *Acetobacter aceti*
5. (5) *Aspergillus oryzae*

Procedure

Observe the permanent slides of these important microbes.

Observation Table

(1) *Streptococcus thermophilus*



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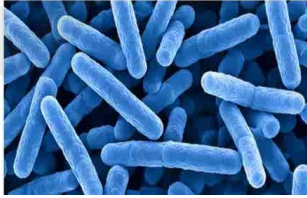
Underline the correct alternative:

Streptococcus thermophilus is a Gram positive/~~Gram negative~~ bacteria.

Cross out the wrong word:

1. *S. thermophilus* is one of the most widely used bacteria in the **dairy**/~~leather~~ industry.
2. **Mozzarella**/~~Anni~~ cheese and yogurt are manufactured using this microbe.
3. It turns milk sugar-**lactose**/~~glucose~~ into lactic acid. The increase in lactic acid turns milk into the gel-like structure which is characteristic of **yogurt**/~~paneer~~.

(2) *Lactobacillus acidophilus*



(2) *Lactobacillus acidophilus*

Underline the correct alternative:

Lactobacillus acidophilus is a Gram positive/~~Gram negative~~ bacteria.

Cross out the wrong word:

1. *L. acidophilus* occurs naturally in the human and animal **gastrointestinal**/~~respiratory~~ tract and **mouth**/~~brain~~.
2. Some strains of *L. acidophilus* are **probiotic**/~~corrosive~~ in nature.

Fill in the blanks:

These strains are commercially used in many dairy products, sometimes together with *streptococcus thermophilus* and *Lactobacillus delbrueckii* subsp.

(3) *Saccharomyces cerevisiae*



(3) *Saccharomyces cerevisiae*

Underline the correct alternative:

Saccharomyces/*cerevisiae* *cerevisiae* is a type of yeast/~~cyanobacteria~~.

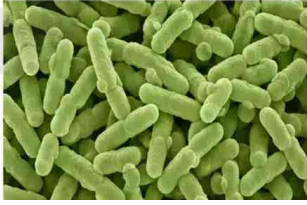
Cross out the wrong word:

1. *S. cerevisiae* used in **wine**/~~cheese~~ making, baking, and brewing since ancient times.
2. It is used as a source of vitamins and minerals. It is rich in **B**/~~D~~ vitamins, ~~chromium~~/**sodium** and **selenium**/~~lead~~. Due to this, health of hair, skin, mouth, liver and **eyes**/~~ears~~ is maintained.

Fill in the blanks:

They also aid in breaking down fats, proteins and carbohydrates for a quick supply of energy.

(4) *Acetobacter aceti*



(4) **Acetobacter aceti**

Underline the correct alternative:

1. *Acetobacter aceti* is a ~~Gram positive~~/Gram negative bacterium.
2. Louis Pasteur/~~Charles Darwin~~ proved that these bacteria cause conversion of ethanol to acetic acid in 1864.

Cross out the wrong word:

1. *A. aceti* is a bacterium that moves using its ~~pseudopodia~~/**peritrichous flagella**. It is present everywhere in the environment, existing in ecological niches which include flowers, fruits, and honey bees, as well as in water and soil.
2. It lives wherever sugar ~~translocation~~/**fermentation** occurs.
3. *A. aceti* is used for the mass production of ~~sulphuric acid~~/**acetic acid** for manufacturing **vinegar**/~~oil~~.
4. It is used for action on **wines**/~~cheese~~ and **ciders**/~~juices~~.

(5) *Aspergillus oryzae*



(5) **Aspergillus oryzae**

Underline the correct alternative:

Aspergillus oryzae is filamentous fungus/~~algae~~.

Cross out the wrong word:

1. *A. oryzae* is widely used in Chinese and other **East Asian**/~~American~~ cuisine.
2. It is used in fermentation of **soybeans**/~~sugar cane~~ for making soya sauce and fermented bean paste.
3. Also used to saccharify **rice**/~~wheat~~, other grains, and potatoes in the making of alcoholic **beverages**/~~vinegar~~.
4. *A. oryzae* is also used for the production of rice **alcohol**/~~vinegars~~.

Inference

When microbiology is used for commercial purpose as well as for economic, social and environment related processes, then it is called industrial microbiology. By sustained research on various microbes, various products are produced on large scale by the industrial microbiology.

Multiple Choice Questions

1. Which of the following microbe is commonly used in baking and brewing?

- (A) *Streptococcus thermophilus*
- (B) *Saccharomyces cerevisiae*
- (C) *Acetobacter*
- (D) *Aspergillus oryzae*

Answer: (B) *Saccharomyces cerevisiae*

2. Which of the following is the microbe used in manufacturing vinegar?

- (A) *Streptococcus thermophilus*
- (B) *Saccharomyces cerevisiae*
- (C) *Acetobacter*
- (D) *Aspergillus oryzae*

Answer: (C) *Acetobacter*

3. Which of the following is the microbe used in manufacturing soya sauce?

- (A) *Streptococcus thermophilus*
- (B) *Saccharomyces cerevisiae*
- (C) *Acetobacter*
- (D) *Aspergillus oryzae*

Answer: (D) *Aspergillus oryzae*

Remember

Various food items are produced with the help of fermentation process. For Example, Bread, Cheese, Wine, etc. Microbes are also used in production of enzymes, medicines, biofuel, pollution control and sewage management.

Date: _____

Teacher's Signature: _____